

## MST-Technologies and Processes

February, 17<sup>th</sup>, 2025

Duration: 120 min.

### Please note

- write your name and matriculation number into the caption of every sheet
- use and write only on the front of the sheet
- use a separate sheet for every task
- write with a pen NOT with a pencil
- if you hinder the exam you will be removed from the lecture hall and fail the exam (5.0 with the statement "Störung")

### Cheating

- any attempt of cheating results in immediate failure of the exam (5.0 with the statement "Täuschung")
- switch off your cell phones and store them in your backpack – a cell phone on the table will be treated as an attempt to cheating

### Withdrawal or absence

- you may withdraw until the moment the tasks are revealed
  - in case you feel unwell during the exam, you may withdraw during the exam for medical reasons; you need to provide (within 3 days) a written note from your doctor confirming the withdrawal for medical reasons
  - please see:  
<https://www.tf.uni-freiburg.de/en/studies-and-teaching/a-to-z-study-faq/de-registration-of-exams>
  - you find the relevant forms here:  
<https://www.tf.uni-freiburg.de/de/studium-lehre/a-bis-z-studium/dokumente/WithdrawalfromExamsduetoIllnessMasterofScience.pdf>
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**Necessary Points:** 50 of 100 in total

**Points:** \_\_\_\_\_

**Result:** \_\_\_\_\_

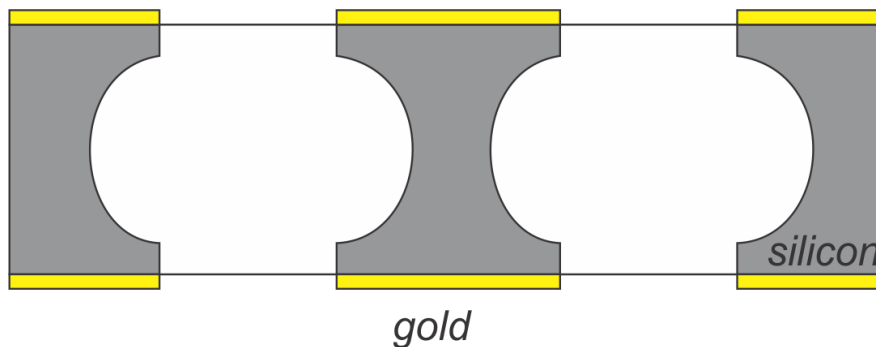


## 1. Processes

24

The following structure is a so-called “double paddle oscillator”. Essentially, the center structure is a freely movable component, the outer two beams are fixed. We ignore the exact fixation of these outer structures in the following and focus only on the manufacturing process to make this structure.

Design the manufacturing steps required to fabricate this structure. Sketch and briefly explain each step. Each manufacturing process (e.g., lithography) used should be depicted out entirely when you first use it. If you use the same process again, you do not need to redraw the entire process.



## 2. Cleanroom Technology

14

- |     |                                                                                                    |   |
|-----|----------------------------------------------------------------------------------------------------|---|
| 2.1 | Name four general recommendations for working in the cleanroom                                     | 4 |
| 2.2 | Sketch and explain the four filtering effects used in Filter Fan Units (FFUs)                      | 4 |
| 2.3 | What are common ISO classes used for white areas in a cleanroom for MEMS and for microelectronics? | 4 |
| 2.4 | What are the two common cleanroom setups? State the name.                                          | 2 |

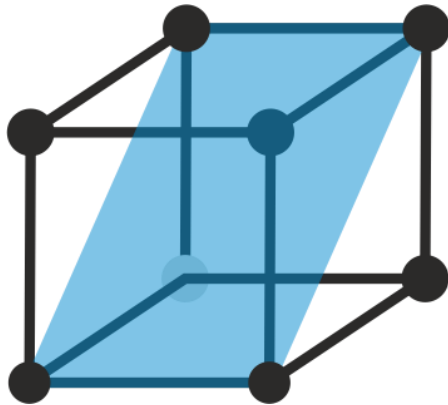


### 3. MEMS Materials

24

3.1 Indicate the following crystal plane

4



3.2 Sketch and explain the Czochralski process for making single-crystalline silicon

8

3.3 Which types of 3-dimensional defects can we have in crystalline materials? Give a brief explanation to every potential defect type

4

3.4 Sketch and explain one of the two processes we discussed for generating Silicon-on-Insulator (SOI) wafers

8

### 4. Thermal oxidation of silicon

14

4.1 Sketch and explain the setup used for pyrogenic wet chemical etching

4

4.2 Dependency on time

(6)

Which are the two stages during thermal oxidation of silicon?

2

Which is the limiting physical quantity in each of these stage?

2

What is the correlation between layer height and time in each of these stages?

2

4.3 Does doping increase or decrease the oxidation rate and why? How is this process called?

4

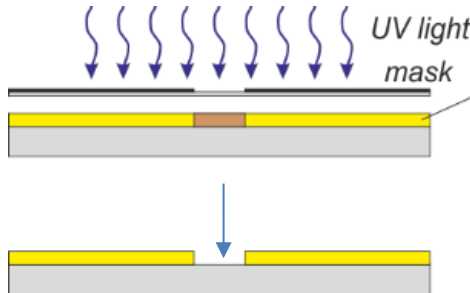


## 5. Photolithography

24

- 5.1 The following image depicts two steps in a lithography workflow: the exposure (top) and the development step (below). Which type of photoresist was used here (what should the missing label read)?

3



- 5.2 Photoresists based on epoxy chemistry usually require a base oligomer/polymer and an additional ingredient which makes them light-responsive. What is this component?
- 5.3 Spin-coating proceeds via four distinct phases. Sketch and explain briefly these four phases.
- 5.4 What is the reason for performing a post-exposure bake?
- 5.5 What are the two parameters that a mask can modulate on the incoming light?
- 5.6 Sketch the lithographic setup in projection lithography. Make sure to include all relevant labels in your sketch.
- 5.7 During characterization of an old bottle of photoresist via an exposure study, you obtain the following graph. Label the characteristic  $D_0$  and  $D_{100}$  values and state the type of the photoresist.

2

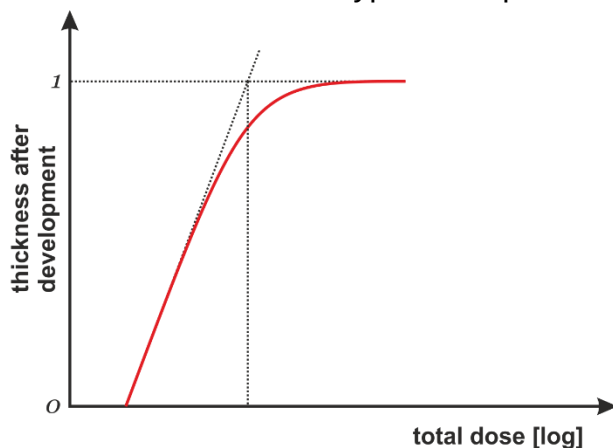
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2

2

4

3



- 5.8 Sketch the classical lithography workflow. Make sure to include all relevant labels.

4

